Calculations and Formulas Reference

Version 2.2.0 | Issued 2026-11-13

Particle Size Distribution (Sieve Analysis)

BS 1377-2:1990, clauses 9.2 / 9.3

Percentage retained and cumulative passing

```
total = SUM(m_i) over every sieve row including the pan

% retained_i = m_i / total * 100

cumulative passing_i = (1 - SUM(retained to row i) / total) * 100
```

Where:

- m_i -- mass retained on sieve row i (g)
- total -- sum of all retained masses (g)

Note: The pan rows (<0.063, <0.075) contribute their mass to the total but report no passing value.

Note: With no weighed material every percentage reads as auto, never 0%.

Note: Worked (lab sheet KIRIAINI AHP): 40.22 g retained on No. 4 of 176.4 g total -> 22.8% retained, 77.2% passing.

`Implemented in psd-sieve-formulas: calculatePercentRetained, calculateCumulativePassing`

Effective sizes D10, D30, D60 (log interpolation)

```
r = (p - P1) / (P2 - P1)

D(p) = 10 ^ (log10(d1) + r * (log10(d2) - log10(d1)))

Cu = D60 / D10

Cc = D30^2 / (D10 * D60)
```

Where:

- p -- target percent passing (10, 30 or 60)
- d1, P1 / d2, P2 -- bracketing sieve size (mm) and passing (%)
- Cu -- coefficient of uniformity
- Cc -- coefficient of curvature

Note: Interpolation runs on the log10 particle-size axis, matching semi-log graph paper.

Note: No D-value is reported when the target falls outside the measured curve.

`Implemented in psd-sieve-formulas: calculateGradationCoeffs`

Full PSD pipeline (processSieveAnalysis)

```
1. Sort sieve rows largest to smallest

2. % retained_i = m_i / total * 100

3. cumulative passing_i = (1 - SUM(retained to row i) / total) * 100

4. D10, D30, D60 via log interpolation; Cu = D60 / D10; Cc = D30^2 / (D10 * D60)

5. Atterberg limits (LL, PL, PI) from sieve-generated data

6. Soil classification: USCS/AASHTO from fractions + Atterberg
```

Where:

- m_i -- mass retained on sieve row i (g)
- total -- sum of all retained masses (g)

Note: Oven-dried LL input: when an oven-dried liquid limit is entered it takes precedence over the air-dried value for classification.

Note: Non-plastic soils (NP) report PI = 0 and skip BS/USCS Atterberg classification.

`Implemented in psd-sieve-formulas: processSieveAnalysis`

Moisture content

```
w = (wet - dry) / dry * 100  (%)
```

Where:

- wet -- wet mass of soil (g)
- dry -- oven-dry mass of soil (g)

`Implemented in gradingCalculations: calculateMoisture`

Hydrometer analysis (Stokes law)

```
Rd = Rn' - Ro'   (reading less the dispersant blank)

Rh = Rd + Cm + Ct   (true reading; Cm = 0.1 g/L meniscus default, Ct = manual temperature correction, default 0)

H = 16.294964 - 0.164 * Rh   (152H)  or  16.294964 - 0.2645 * Rh   (151H)   [cm]

K = 1000 * SQRT(18 * nu * 10^-6 / g)

D = K * SQRT((H / 100) / ((Gs - 1) * t * 60))   [mm]

K% = Rh * (Gs / (Gs - 1)) * V / (m * 1000) * 100   [% finer]
```

Where:

- Rn' -- observed hydrometer reading (g/L)
- Ro' -- dispersant (zero) correction reading (g/L)
- H -- effective depth of the hydrometer bulb (cm)
- nu -- kinematic viscosity of water, from Table 7 by temperature (15-32 C, clamped; default 1.005)
- g -- 9.80665 m/s^2
- t -- elapsed time (min)
- Gs -- specific gravity of solids (default 2.65)
- V -- suspension volume (cm3, default 1000)
- m -- soil mass: hydrometer subsample for K (sample), whole-sample mass for K (whole)

Note: BS 1377-2:1990 9.5.7.2: the 1990 edition carries no temperature-correction term; a manual value is accepted for correction-based instruments.

Note: The (Gs - 1) divisor converts the density excess on the hydrometer scale into mass of soil in suspension, so K% needs Gs > 1.

Note: Sheet columns map to code fields: Rn' -> observed reading, Rd -> adjusted reading, Rh -> corrected reading, H -> effective depth, D -> particle diameter, K% sample/whole -> fines in suspension / by hydrometer.

`Implemented in gradingCalculations: calculateHydrometer`

Worked example - Group Index (AASHTO M 145 6.4)

```
GI = (F - 35) * (0.2 + 0.005 * (LL - 40)) + 0.01 * (F - 15) * (PI - 10)

F = 82, LL = 38, PI = 21:

GI = (82 - 35) * (0.2 + 0.005 * (38 - 40)) + 0.01 * (82 - 15) * (21 - 10)

GI = 8.9 + 7.4 = 16.3 -> reported as the whole number 16
```

Where:

- F -- % passing the No. 200 (0.075 mm) sieve

Note: The standard's own worked example; negative results are reported as zero.

Note: A-2-6 and A-2-7 use the plasticity term alone.

`Implemented in soilClassification: calculateAashtoGroupIndex`

Atterberg Limits

BS 1377-2:1990, clause 4.3 (cone penetrometer) / ISO 17892-12

Trial moisture content from masses

```
water = wet - dry

dry soil = dry - tare
```

```
w = water / dry soil * 100  (%)
```

Where:

- wet -- container + wet soil (g)
- dry -- container + oven-dry soil (g)
- tare -- empty container (g)

Note: Withheld (blank) when dry soil <= 0 or water < 0.

Note: A directly entered moisture is used only when no masses are available.

`Implemented in atterbergCalculations: calculateMoistureFromMass, getTrialMoisture, getWaterMass, getDrySoilMass`

Liquid Limit (cone penetrometer, 20 mm)

```
LL = moisture of any valid trial at exactly 20 mm penetration

otherwise fit w = a * log10(penetration) + b by least squares and read w at 20 mm
```

Where:

- w -- trial moisture content (%)
- penetration -- cone penetration (mm), working range 10-30

Note: Fewer than two usable points (and none at 20 mm) yields no result.

Note: This is the cone method; the Casagrande cup (blow counts, ASTM D4318) is a different procedure.

`Implemented in atterbergCalculations: calculateLiquidLimit, calculateLogLinearRegression, LL_TARGET_PENETRATION_MM, LL_MIN_VALID_TRIALS`

Plastic Limit

```
PL = mean of the valid plastic-limit trial moistures
```

Note: At least two valid trials are required.

`Implemented in atterbergCalculations: calculatePlasticLimit, PL_MIN_VALID_TRIALS`

Linear shrinkage

```
LS = mean of (L0 - L) / L0 * 100  (%)
```

Where:

- L0 -- initial mould length (mm)
- L -- oven-dry length (mm)

Note: This is linear shrinkage from length change; it is NOT the BS 1377-2 4.5 shrinkage limit, which is determined differently.

`Implemented in atterbergCalculations: calculateLinearShrinkage, LS_MIN_VALID_TRIALS`

Non-plastic soils (NP)

```
PI = LL - PL = 0 (PL entered equal to LL, or the Non-plastic box ticked)

-> code NP: BS/USCS classification is skipped, soil reported as Non-plastic
```

Note: Per lab note: when shrinkage reads 0.00 and PI is zero with no plastic-limit entries, display Non-plastic rather than classifying.

`Implemented in atterbergCalculations: classifyAtterberg`

Plasticity Index and related indices

```
PI = LL - PL  (withheld when PL exceeds LL: physically invalid)

Modulus of plasticity = PI * (% passing 425 um)

A-line: PI = 0.73 * (LL - 20)

U-line: PI = 0.90 * (LL - 8)  (upper bound for natural soils)
```

Where:

- PI -- plasticity index (%)

Note: At or above the A-line the fines classify as clay (C); below it as silt (M); LL = 50 splits low (L) from high (H) plasticity.

Note: PI above the U-line is flagged as suspect data and classification is withheld.

`Implemented in atterbergCalculations: calculatePlasticityIndex, calculateModulusOfPlasticity, getALinePI, getULinePI, classifyAtterberg`

Worked example - classification (programmer's guide)

```
LL = 48, PL = 22 -> PI = 48 - 22 = 26

A-line value = 0.73 * (48 - 20) = 20.44; PI 26 >= 20.44 -> Clay (C)

LL 48 in [35, 50) -> Intermediate (I); BS code = C + I = CI

USCS: LL < 50 and above A-line -> CL

Description: Clay of Intermediate Plasticity; note: Moderate volume change potential
```

Note: The guide's own example; PI is always derived, never entered directly.

`Implemented in atterbergCalculations: classifyAtterberg, getALinePI`

Plasticity Index

```
PI = LL - PL
```

Where:

- PI -- plasticity index (%)
- LL -- liquid limit (%)
- PL -- plastic limit (%)

Note: Per Excel formula from ATTERTEST.xlsx: PI = LL - PL.

Note: The psd-sieve-formulas module implements calculatePlasticityIndex(ll, pl).

`Implemented in psd-sieve-formulas: calculatePlasticityIndex`

Liquidity Index

```
LI = (w - PL) / PI
```

Where:

- LI -- liquidity index
- w -- natural water content (%)
- PL -- plastic limit (%)
- PI -- plasticity index (LL - PL)

Note: Per Excel formula from ATTERTEST.xlsx: LI = (w - PL) / PI.

Note: The psd-sieve-formulas module implements calculateLiquidityIndex(w, pl, pi).

`Implemented in psd-sieve-formulas: calculateLiquidityIndex`

Soil Classification by Atterberg Limits

```
A-line method: PI = 0.73 * (LL - 20); classify as CL/CH/MH/ML/CL-ML
```

Where:

- LL -- liquid limit (%)
- PI -- plasticity index (%)
- fines -- fines content % passing #200 sieve

Note: Per Excel formula from ATTERTEST.xlsx: A-line classification using psd-sieve-formulas.classifySoilByAtterberg().

Note: The psd-sieve-formulas module implements classifySoilByAtterberg(ll, pi, fines).

Note: Returns CL (lean clay), CH (fat clay), ML (silt), MH (elastic silt), or CL-ML (silty clay).

`Implemented in psd-sieve-formulas: classifySoilByAtterberg`

Soil Classification (BS, USCS, AASHTO)

BS 1377 / ASTM D2487 (USCS) / AASHTO M 145 - ASTM D3282

BS 1377 and USCS decision rules (from LL and PL)

```
PI = LL - PL (always derived, never entered directly)

A-line value = 0.73 * (LL - 20): PI >= A-line -> Clay (C), else Silt (M); ties go to Clay

BS second letter: LL < 35 L (Low) / < 50 I (Intermediate) / < 70 H (High) / < 90 V (Very High) / >= 90 E (Extremely High); a boundary belongs to the higher band

BS code = first + second letter (e.g. C + I = CI, Clay of Intermediate Plasticity)

USCS: LL < 50 -> CL / ML, LL >= 50 -> CH / MH on the same A-line split

Hatched zone LL < 50, 4 <= PI <= 7, at/above A-line -> dual symbol CL-ML

Guards: PL > LL invalid; PI = 0 -> NP (skip); PI < 4 borderline; PI > U-line 0.9 * (LL - 8) suspect (withhold)
```

Where:

- LL -- liquid limit (%)
- PL -- plastic limit (%)
- PI -- plasticity index (%)

Note: Engineering notes by band: Low / Moderate / High / Very high volume change potential.

Note: Per the programmer's guide the whole classification is two comparisons and a string join.

`Implemented in atterbergCalculations: classifyAtterberg, getALinePI, getULinePI`

USCS grain-size route (ASTM D2487 flow charts)

```
fines = % passing No. 200 (0.075 mm)

gravel = 100 - % passing No. 4 (4.75 mm)

sand = % passing No. 4 - % passing No. 200

fines >= 50% -> fine-grained (A-line route above); else sand family if sand > gravel, gravel family otherwise

clean (fines <= 12%): SP/SW or GP/GW; fines > 12%: the A-line decides silty (M) vs clayey (C)
```

Note: Gravel + sand + fines must total 100%; otherwise classification is withheld as invalid data.

Note: Relaxed classification gate: fine-grained soils (>= 50% fines) classify on the No. 200 passing value alone.

Note: Manual USCS/AASHTO overrides take precedence over auto-classification on the printed report.

`Implemented in soilClassification: classifySoilUSCS, validateClassificationData`

AASHTO M 145 groups (12 groups, lab classification note)

```
P200 <= 35% -> granular soils, else silt-clay soils

A-1-a: P10 <= 50, P40 <= 30, P200 <= 15, PI <= 6 (stone fragments, gravel, sand)

A-1-b: P40 <= 50, P200 <= 25, PI <= 6 (stone fragments, gravel, sand)

A-3: P40 >= 51, P200 <= 10, non-plastic (fine sand)

A-2-4: P200 <= 35, LL <= 40, PI <= 10 / A-2-5: LL > 40, PI <= 10

A-2-6: P200 <= 35, LL <= 40, PI > 10 / A-2-7: LL > 40, PI > 10

A-4: P200 > 35, LL <= 40, PI <= 10 / A-5: LL > 40, PI <= 10 / A-6: LL <= 40, PI > 10

A-7-5: P200 > 35, LL > 40, PI <= LL - 30 / A-7-6: PI > LL - 30

GI = (F - 35) * (0.2 + 0.005 * (LL - 40)) + 0.01 * (F - 15) * (PI - 10)
```

Where:

- P10 / P40 / P200 -- % passing No. 10 (2.00 mm) / No. 40 (0.425 mm) / No. 200 (0.075 mm)
- F -- % passing No. 200 for the Group Index

- GI -- group index, whole number >= 0 (A-2-6/A-2-7 use the plasticity term alone)

Note: LL and PI are measured on the fraction passing No. 40; LL plays no role in A-1-a, A-1-b or A-3.

Note: Ratings: A-1/A-2/A-3 Excellent-Good (GI 0-4); A-4..A-7 Fair-Poor (GI up to 8/12/16/20).

Note: Critical note: all three sieve values are required for granular soils; without P10/P40 the app shows a best-effort group flagged as unverified.

`Implemented in soilClassification: classifySoilAASHTO, calculateAashtoGroupIndex, getAashtoEvidenceWarnings`

Compaction (Proctor)

BS 1377-4:1990, clause 3.3

Per-point densities and moisture

```
wet soil = mould wet - mould tare

water = container wet - container dry

dry soil = container dry - container tare

bulk density rho = wet soil / V * 1000   (kg/m3, V in cm3)

w = water / dry soil * 100  (%)

dry density rho_d = rho / (1 + w / 100)   (kg/m3)
```

Where:

- V -- mould volume (cm3)
- w -- moisture content (%)

Note: A point with any missing measurement yields no densities rather than a partial result.

Note: The moisture content divides by the container tare: a tin weight entered one column over silently turns a 10% moisture content into several hundred per cent. A container tare of 0 g is legitimate (the recorded masses are then net of tin) and is not flagged.

Note: Entry checks flag a moisture content above 60% (implausible for a compacted soil), a dry density above the particle density, a mould tare of zero, and any point whose masses fail the physical checks - see proctorRowWarnings. These still print in the report.

`Implemented in proctorRecords: calculateProctorPoint, proctorRowWarnings`

Optimum moisture content and maximum dry density

```
fit rho_d = a*w^2 + b*w + c through the three readings around the measured maximum

OMC = -b / (2a),  MDD = a*OMC^2 + b*OMC + c

R^2 = 1 - SUM(residual^2) / SUM((rho_d - mean)^2), checked against all plotted readings
```

Where:

- OMC -- optimum moisture content (%)
- MDD -- maximum dry density (kg/m3)

Note: The fitted compaction curve on screen and in the PDF/xport is a Catmull-Rom spline through every measured point, matching the curves drawn on the laboratory's report sheets.

Note: The three-point parabola through the measured peak and its two neighbours is used only to derive OMC/MDD; the drawn spline is the same curve the technicians draw by hand through all readings.

Note: The measured maximum must be bracketed; the local fit must open downward and its vertex must lie between the neighboring moisture contents.

Note: When no local parabola can be supported, OMC/MDD fall back to the single densest measured point (reported as peak-point basis).

Note: New records default specific gravity Gs to 2.0 and the Y-axis window to 1400-1550 kg/m3; the window is stored per project on the Proctor payload (yAxisMin / yAxisMax), and the exported PDF renders every sample against it.

`Implemented in proctorRecords: fitCompactionCurve, calculateProctor`

Zero air voids and air-voids lines

```
ZAV: rho_d = 1000 * Gs / (1 + (w / 100) * Gs)

air voids na: rho_d(na) = ZAV * (1 - na / 100)
```

Where:

- Gs -- measured specific gravity of soil solids
- na -- target air voids content (%)

Note: Zero air voids points above this line are not currently flagged: the two BS 1377-4 plausibility notices - 'Specific gravity outside the 2.4 to 2.9 range of mineral soils' and 'N points plot above the zero air voids line' - are suppressed in this build by default (SHOW_SPECIFIC_GRAVITY_WARNINGS=false in proctorRecords.ts) per laboratory decision; reprint with the flag enabled once Gs is settled for the granular soils.

Note: The fitted compaction curve must stay below the ZAV line; the air-voids line always sits below ZAV by construction. Enforced non-conformity is currently handled by the laboratory during review, not by the print.

Note: Report sheets draw the 0%, 5% and 10% voids lines beside the fitted curve for the same check.

`Implemented in proctorRecords: zeroAirVoidsDensity, airVoidsDensity`

Compressive Strength of Concrete Cubes

BS EN 12390-3:2002 (testing) / BS EN 206 (conformity)

Compressive strength

```
strength = load_kN * 1000 / (W * H)   (MPa = N/mm2)
```

Where:

- load_kN -- maximum load at failure (kN)
- W * H -- loaded cross-section (mm x mm)

Note: A genuine 0 MPa result is kept distinct from an unfilled field: blanks stay blank, never 0.

Note: Negative loads and non-positive dimensions yield no strength.

`Implemented in compressiveCalculations: strengthOf, parseNumber`

Bulk density and age

```
density = (mass / 1000) / ((W * H * D) / 10^9)   (kg/m3)

age = floored whole days from date of cast to date of test
```

Where:

- mass -- cube mass (g)
- W * H * D -- cube volume (mm3)

Note: Densities outside 2200-2600 kg/m3 are flagged as abnormal on screen and in charts.

Note: A test date before the cast date yields no age rather than a negative one.

`Implemented in compressiveCalculations: densityOf, ageOf`

Target strength and age groups

```
class target: C25/30 -> 30 (the cube figure); single-figure classes use that figure

bands: 1 day exact, 3 days exact, 7 +/- 1, 14 +/- 2, 28 +/- 3 days

band targets follow the class through the gain table (C30 -> 4.8 / 12 / 19.5 / 27 / 29.7
MPa) until typed over
```

Note: Each band is judged on its own mean: cubes of different ages are never averaged together.

Note: Cubes outside every band are reported under Other ages, never dropped.

Note: Early/late breaks off the nominal day fall into Other ages; confirm wider windows with the lab if needed.

`Implemented in compressiveCalculations: cubeStrengthFromClass, buildAgeGroups, classBandTargets, STRENGTH_GAIN_TABLE`

Strength gain with age (lab reference table)

```
1 day -> 16% of 28-day strength

3 days -> 40% of 28-day strength

7 days -> 65% of 28-day strength
```

```
14 days -> 90% of 28-day strength

28 days -> 99% of 28-day strength

expected MPa = class target * percent / 100

between table ages the expectation interpolates linearly (10 days -> 75.7%); past 28 days
it holds 99%; below 1 day there is none
```

Note: Percentages are of the characteristic (28-day) strength, e.g. C30 at 7 days is expected to reach 19.5 MPa.

Note: Report sheets print ages past 31 days as Above 28 instead of the exact day count.

`Implemented in compressiveCalculations: expectedPercentAtAge, expectedStrength, formatReportAge`

Acceptance

```
group accepted when mean >= target AND lowest cube >= target - 4

per cube: Satisfactory when strength reaches the age expectation (e.g. >= 65% of class at
7 days, >= 99% at 28 days); unknown age falls back to >= 65%

pass tally: cubes with strength >= threshold (a real 0 MPa counts as a failure)
```

Where:

- target -- band target strength (MPa)

Note: The simple pass/fail tally is indicative only; conformity is judged on the group verdict above.

Note: Strength bands: < 7 very low, 7-20 low, 20-40 normal structural, > 40 high.

`Implemented in compressiveCalculations: groupVerdict, getPassFailResults, strengthCategory, gainSummary, ACCEPTANCE_MARGIN_MPA`

Report percentages and remarks (lab cube sheet)

```
% of class = strength / class target * 100 (whole number)

Satisfactory when strength reaches the age expectation from the gain table (65% at 7 days,
99% at 28 days)
```

Note: Lab sheet VENUS: 25.1 MPa of C30 -> 84% Satisfactory; 21.5 MPa -> 72% Satisfactory.

`Implemented in compressivePdfGenerator: percentOfClass, isSatisfactory`

Worked example - 150 mm cube

```
load 564.1 kN on 150 x 150 mm: strength = 564.1 * 1000 / (150 * 150) = 25.07 MPa (lab
sheet rounds to 25.1)

mass 7927 g: density = (7927 / 1000) / ((150 * 150 * 150) / 10^9) = 2349 kg/m3

cast 19-Sep, tested 26-Sep -> age 7 days -> 7-day band (7 +/- 1)
```

Note: The lab sheet's own readings; recompute them to verify any result row.

`Implemented in compressiveCalculations: strengthOf, densityOf, ageOf`

California Bearing Ratio (CBR)

BS 1377-4:1990 / ASTM D1883

CBR at 2.5 mm and 5.0 mm penetration

```
CBR_2.5 = (load_2.5 / 13.24) * 100  (%)

CBR_5.0 = (load_5.0 / 19.96) * 100  (%)

reported CBR = max(CBR_2.5, CBR_5.0)
```

Where:

- load_2.5 -- load at 2.5 mm penetration (kN)
- load_5.0 -- load at 5.0 mm penetration (kN)
- 13.24 -- standard load at 2.5 mm (kN)

- 19.96 -- standard load at 5.0 mm (kN)

Note: Standard loads are 13.24 kN at 2.5 mm and 19.96 kN at 5.0 mm penetration.

Note: The reported CBR is the greater of the two values.

`Implemented in cbrCalculations: calculateCBR, calculateCBRRows, reportCBR, STANDARD_LOAD_2_5, STANDARD_LOAD_5_0`

Direct Shear Test

BS 1377-7:1990 / ASTM D3080

Mohr-Coulomb failure envelope

```
tau = c + sigma * tan(phi)

slope = (n * SUM(x*y) - SUM(x) * SUM(y)) / (n * SUM(x^2) - SUM(x)^2)

intercept = (SUM(y) - slope * SUM(x)) / n

phi = atan(slope) * 180 / PI  (degrees)

c = intercept  (cohesion, kPa)
```

Where:

- tau -- shear stress at failure (kPa)
- c -- cohesion / intercept of failure envelope (kPa)
- sigma -- normal stress (kPa)
- phi -- angle of internal friction (degrees)
- x -- normal stress data points
- y -- shear stress data points
- n -- number of shear boxes

Note: Requires at least two shear boxes with different normal stresses.

*Note: Linear regression fits tau = c + sigma * tan(phi) to the (normal, shear) pairs.*

Note: The slope gives tan(phi); the intercept gives cohesion c.

`Implemented in shearCalculations: calculateShearEnvelope, shearStrength`

Consolidation Test

BS 1377-6:1990 / ASTM D2435

Time-settlement relationship

```
settlement_i = reading_i  (mm)

max_settlement = MAX(settlement_1 ... settlement_n)
```

Where:

- settlement_i -- settlement reading at time i (mm)
- time_i -- elapsed time since load application (min)
- max_settlement -- maximum settlement observed (mm)

Note: Plots settlement vs time to determine consolidation behavior.

Note: Maximum settlement is the largest recorded settlement value.

`Implemented in consolidationCalculations: calculateConsolidationPoints, maxSettlement`

Unconfined Compressive Strength (UCS)

ISRM SM (2007) / ASTM D7012-14

Uniaxial compressive strength

```
qu = 4F / (pi d^2)   (MPa = N/mm^2)

A = (pi / 4) * d^2   (mm^2)

mean, sample standard deviation, coefficient of variation (n >= 3)
```

Where:

- qu -- uniaxial compressive strength (MPa)

- F -- maximum load at failure (kN)
- d -- specimen diameter (mm)
- L/D -- length-to-diameter ratio; must lie between 2 and 3 (ISRM SM 2007)
- A -- cross-sectional area of specimen (mm^2)

Note: ISRM SM 2007 requires at least 3 valid specimens; cylindrical specimens with L/D between 2 and 3; flat, parallel, polished ends; loading rate 0.5 to 1.0 MPa/s.

Note: Failures through the specimen ends are rejected and the specimen must be re-tested.

Note: Report mean, standard deviation and coefficient of variation. Strength alone is not reported without the number of specimens.

Note: The test is repeated for at least 3 valid specimens; a result is not reportable unless the set conforms ISRM SM 2007 / ASTM D7012-14.

`Implemented in ucsCalculations: calculateSpecimen, sampleStandardDeviation, assessConformity, mean, MIN_SPECIMENS, STRESS_RATE_MIN, STRESS_RATE_MAX, L_D_MIN, L_D_MAX`

Point Load Test

ISRM SM (2007) / ASTM D5731-14

Point load strength index

```
Is = P / De^2   (MPa), with P in N and De in mm

Is(50) = Is * (De / 50)^0.75   (normalised to a 50 mm reference diameter)

mean, sample standard deviation, coefficient of variation (n >= 10)
```

Where:

- Is(50) -- normalised point load strength index for a 50 mm reference diameter (MPa)
- Is -- unnormalised point load index (MPa)
- P -- load at failure (kN)
- De -- equivalent specimen diameter (mm); De = D for diametral tests, sqrt(4A/pi) for lumps/blocks; test close to 50 mm preferred
- COV -- coefficient of variation of the sample; > 40% is rejected by ISRM SM

Note: ISRM SM 2007 recommends at least 10 valid tests per set; De close to 50 mm is preferred; De outside 25-75 mm is flagged.

Note: Failure should preferably be a single fracture through the mid-section; through-end failures are rejected and the specimen re-tested.

Note: Ends fail when load is applied by two conical platens; a loading rate giving failure within approximately one minute is recommended.

Note: Results may be compared to unconfined compressive strength only through material-class-dependent correlations.

`Implemented in pointLoadCalculations: calculatePointLoadIndex, calculateSpecimen, sampleStandardDeviation, assessConformity, calculatePointLoadRows, averagePointLoadIndex, MIN_SPECIMENS, DE_MIN_MM, DE_MAX_MM, COV_MAX, IS50_CORRECTION_EXPONENT`

Porosity Test

BS 1377-2:1990

Porosity from mass and volume

```
n = ((W_sat - W_dry) / V) * 100  (%)
```

Where:

- n -- porosity (%)
- W_sat -- saturated (wet) mass of sample (g)
- W_dry -- oven-dry mass of sample (g)
- V -- volume of sample (cm^3)

Note: (W_sat - W_dry) is the mass of water in the pores; numerically equal to pore volume in cm^3 (water density = 1 g/cm^3).

Note: Requires dry weight, saturated weight and volume all to be positive.

`Implemented in porosityCalculations: calculatePorosity, calculatePorosityResult`

Ultrasonic Pulse Velocity Test (UPVT)

BS EN 12504-4:2004 / ASTM C597

Pulse velocity

```
V = L / T   (km/s)
```

Where:

- V -- ultrasonic pulse velocity (km/s)
- L -- path length between transducers (mm)
- T -- transit time of the pulse (microseconds)

Note: 1 mm / 1 microsecond = 1 km/s, so V = L / T directly.

Note: Quality classification: >=4.5 Excellent, >=3.5 Good, >=3.0 Medium, >=2.0 Poor, <2.0 Very Poor.

`Implemented in upvtCalculations: calculateVelocity, calculateUPVTRows, averageVelocity, qualityFromVelocity`

Schmidt Hammer / Rebound Hammer

BS EN 12504-2:2001 / ASTM C805

Rebound number and estimated strength

```
R_avg = SUM(retained readings) / retained count

reading differs by more than 12% from mean -> discarded; more than 2 rejected -> area invalid

f_c = M * R_avg + B   (only when a site-specific correlation M/B is configured in the record)
```

Where:

- R_avg -- average rebound number of the retained readings
- f_c -- estimated compressive strength (MPa)
- M -- site correlation slope (MPa per rebound)
- B -- site correlation intercept (MPa)

Note: Ten rebound readings per test area are required by BS EN 12504-2 / ASTM C805 / BS 1881-202.

Note: Any reading varying from the mean by more than 12% is discarded; if more than two are discarded, the area is unsuitable and must be re-taken.

Note: The strength correlation is never assumed. If the record carries no correlation, the report prints rebound numbers only; estimated strength is printed only with a configured site-specific correlation.

Note: Anvil calibration, impact orientation and surface condition must be documented for a conforming (completed) test; the print states conformity or lists what is missing.

`Implemented in schmidtCalculations: getValidReadings, evaluateSchmidtRow, calculateAverageRebound, estimateStrengthFromRebound, calculateSchmidtRows, averageRebound, rowsConform, REQUIRED_READINGS`

Schmidt Hammer (Rock)

ISRM SM (1978) Rebound (Schmidt) hammer testing of rocks

Rebound hardness of rock

```
R_avg = SUM(retained readings) / retained count

reading differs by more than 12% from mean -> discarded; more than 2 rejected -> area invalid
```

Where:

- R_avg -- average rebound number of the retained readings

Note: ISRM SM (1978) requires at least 10 rebound readings per test area, equally spaced and away from fractures, edges and weathered surfaces.

Note: Any reading varying from the mean by more than 12% is discarded; if more than two are discarded, the area is unsuitable and must be re-taken.

Note: The rebound number is a surface-hardness index. It is NOT a stand-in for compressive strength; UCS must be determined by laboratory tests on representative specimens. No MPa correlation is printed for rock.

Note: Anvil calibration and impact orientation must be documented for a conforming (completed) test; the print states conformity or lists what is missing.

`Implemented in schmidtCalculations: getValidReadings, evaluateSchmidtRow, calculateAverageRebound, calculateSchmidtRows, averageRebound, rowsConform, REQUIRED_READINGS`

Concrete Coring Test

BS EN 12504-1:2009

Core strength and L/D correction

```
Area = pi * (d / 2)^2   (mm^2; reported in m^2 as Area / 10^6)

Volume = Area * length   (mm^3; reported in m^3 as Volume / 10^9)

Density = (weight in air / 1000) / (Volume in m^3)   (kg/m^3)

strength = (P * 1000) / Area   (MPa)

L/D = length / diameter

factor = 1.0  when L/D >= 2.0

factor = 0.87 + 0.065 * (L/D)  when L/D < 2.0

corrected = strength * factor
```

Where:

- d -- core diameter (mm)
- length -- core length after trimming (mm)
- P -- failure load (kN)
- Area -- loaded cross-sectional area (mm^2)
- L/D -- length-to-diameter ratio
- corrected -- L/D-corrected core strength (MPa)

Note: Cores with L/D < 2.0 are strength-reduced: the factor is below 1.0, so the corrected strength is lower than the core strength.

Note: Cores with L/D >= 2.0 use the uncorrected strength.

Note: Average corrected strength is reported across all cores.

`Implemented in coringCalculations: calculateLDRatio, calculateCoreStrength, lDCorrectionFactor, calculateCorrectedStrength, calculateCoringRows, averageCorrectedStrength`

Cone Penetration Test

ISO 22476-1 / Robertson (1986) simplified SBT chart

Friction ratio Rf

```
Rf = fs / qc * 100   (%)

qc is entered in MPa and fs in kPa, so Rf = fs / (qc * 10)
```

Where:

- fs -- sleeve friction (kPa)
- qc -- cone resistance (MPa)
- Rf -- friction ratio (%)

Note: Rf is derived, never typed: it is recomputed from qc and fs on edit and on load.

Note: A zero sleeve friction is legitimate (clean sand) and gives Rf = 0.

Note: Typical Rf range: about 0.5 to 8 percent; high Rf means cohesive soil.

`Implemented in cptCalculations: calculateFrictionRatio, rederiveCptRow`

Soil classification (Robertson simplified chart)

```
Rf >= 8                      -> Clay

Rf >= 5                      -> Silty Clay to Clay
```

```
qc > 10,  Rf < 1              -> Sand
qc > 10,  Rf >= 1             -> Very Stiff Sand to Clayey Sand
qc 5-10,  Rf < 1.5            -> Sand to Silty Sand
qc 5-10,  Rf 1.5-5            -> Silty Sand to Sandy Silt
qc 2-5,   Rf < 2              -> Silty Sand to Sandy Silt
qc 2-5,   Rf 2-5              -> Clayey Silt to Silty Clay
qc 1-2,   Rf < 3              -> Sandy Silt to Clayey Silt
qc 1-2,   Rf 3-5              -> Clayey Silt to Silty Clay
qc <= 1,  Rf < 5              -> Clayey Silt to Silty Clay
```

Where:

- qc -- cone resistance (MPa)
- Rf -- friction ratio (%)

Note: The ladder is banded axis-aligned and fully closed: any valid (qc > 0, Rf >= 0) pair classifies.

Note: Earlier banding had gaps (e.g. qc > 10 with Rf >= 1) that returned Unclassified.

`Implemented in cptCalculations: classifySoilRobertson`

Slump Test

BS EN 12350-2:2019 / EN 206 consistence classes / ASTM C143

Slump value

```
Slump = cone height - height of the subsided concrete   (mm)

Standard Abrams cone: 200 mm base, 100 mm top, 300 mm height
```

Where:

- cone height -- mould height, 300 mm standard (mm)
- height after subsidence -- highest point of the slumped concrete (mm)
- Slump -- subsidence of the fresh concrete (mm)

Note: Recorded to the nearest 10 mm under EN 12350-2 and to the nearest 5 mm under ASTM C143.

Note: Only a true slump counts; a shear or collapse slump is discarded and the test repeated.

Note: The slump is derived, never typed: it is recomputed from the heights on edit and on load.

`Implemented in slumpCalculations: calculateSlump, rederiveSlumpEntry`

EN 206 consistence classes

```
S1  10-40 mm

S2  50-90 mm

S3  100-150 mm

S4  160-210 mm

S5  above 210 mm
```

Where:

- Slump -- subsidence of the fresh concrete (mm)

Note: Ranges per BS 8500-1. Intermediate readings round to the nearest class at the midpoints (45, 95, 155, 215) so every slump in the sensitive range classifies.

Note: Below 10 mm is beneath the EN 12350-2 sensitive range and reports Below S1 rather than a class.

`Implemented in slumpCalculations: classifySlumpClass, describeWorkability`

Target tolerance and validity

```
Target slump <= 40 mm:  tolerance +/-30 mm

Target slump 50-90 mm:  tolerance +/-40 mm

Target slump >= 100 mm: tolerance +/-50 mm

EN 12350-2 sensitive range: 10-210 mm
```

Where:

- Target -- specified slump (mm)
- Slump -- measured slump (mm)

Note: Tolerances on a spot sample per BS 8500-1 basis (Concrete Society); targets are optional and their absence is not an error.

Note: Above 210 mm the cone test is unsuitable (collapse likely) - use a slump flow (SCC) test instead.

`Implemented in slumpCalculations: toleranceBand, assessCompliance, assessValidity`